

17	resistors in series	$R_T = R_1 + R_2$
18	resistors in parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
19	capacitors	time constant : $\tau = RC$
20	Lorentz factor	$\gamma = \frac{1}{\sqrt{1 - v^2 / c^2}}$
21	time dilation	$t = t_0 \gamma$
22	length contraction	$L = L_0 / \gamma$
23	relativistic mass	$m = m_0 \gamma$
24	total energy	$E_{total} = E_k + E_{rest} = mc^2$
25	universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
26	mass of Earth	$M_E = 5.98 \times 10^{24} \text{ kg}$
27	radius of Earth	$R_E = 6.37 \times 10^6 \text{ m}$
28	mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
29	charge on the electron	$e = -1.6 \times 10^{-19} \text{ C}$
30	speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$

Prefixes/Units

$$\text{p} = \text{pico} = 10^{-12}$$

$$\text{n} = \text{nano} = 10^{-9}$$

$$\mu = \text{micro} = 10^{-6}$$

$$\text{m} = \text{milli} = 10^{-3}$$

$$\text{k} = \text{kilo} = 10^3$$

$$\text{M} = \text{mega} = 10^6$$

$$\text{G} = \text{giga} = 10^9$$

$$\text{t} = \text{tonne} = 10^3 \text{ kg}$$

END OF FORMULA SHEET